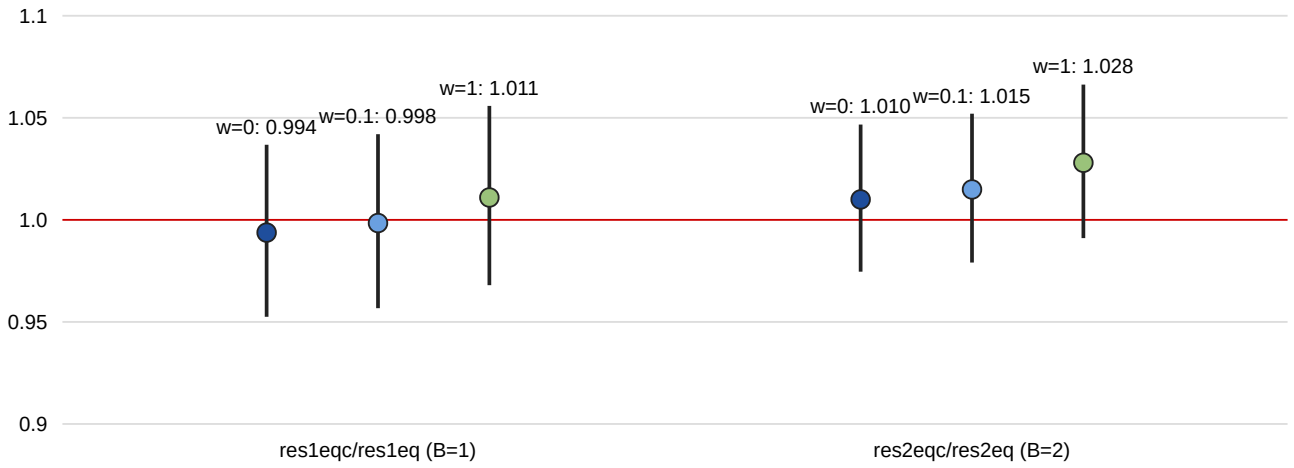
res1eq engineering step: chained restarts. Null result.

Class: engineering, not met (C1 fails). Goal (IC near rho parity) is not met: res1eq is 1.60 × matched rho at w=0, 1.80 × at w=0.1 and 2.70 × at w=1 (harness band, this run).

Change: `residual-fused-eq-chain` restarts each walk at *previous start + random jump* (one addition) instead of two scalar multiplications. Prediction (PROTOCOL.md, frozen before data): at most about 3% saving, since restarts were ≈2.6% of cost and the exact two-relation collision floor is 1.5 × rho (res1eq is within about 6% of it).

Result (C1)



paired ratio of counted cost, candidate/baseline, with 95% interval; 256 targets x 2 rounds, p36. Below 1.0 = candidate cheaper.

paired ratio	w=0	w=0.1	w=1
res1eqc/res1eq (B=1)	0.9938 [0.9525, 1.0368]	0.9984 [0.9567, 1.0420]	1.0110 [0.9680, 1.0558]
res2eqc/res2eq (B=2)	1.0100 [0.9746, 1.0467]	1.0149 [0.9791, 1.0520]	1.0280 [0.9911, 1.0663]

C1 needs the upper 95% bound below 1 at all three prices. B=1: upper bounds 1.037, 1.042, 1.056; B=2: 1.047, 1.052, 1.066. **C1 not met**; the point estimates (0.994 at w=0, 1.011 at w=1 for B=1) are inside the noise of 256 paired targets (about ±4%). The predicted ≈0.97 is not distinguishable from 1 and not supported.

Distance to matched rho (C2)

ratio	w=0	w=0.1	w=1
rho→res1eq	1.6011 [1.5185, 1.6882]	1.8000 [1.7069, 1.8981]	2.6958 [2.5556, 2.8438]
rho→res1eqc	1.5911 [1.5123, 1.6740]	1.7972 [1.7080, 1.8911]	2.7254 [2.5893, 2.8687]
rho→res2eq	2.0236 [1.9211, 2.1316]	2.2744 [2.1590, 2.3959]	3.4039 [3.2303, 3.5867]
rho→res2eqc	2.0439 [1.9430, 2.1500]	2.3083 [2.1941, 2.4284]	3.4991 [3.3252, 3.6821]

res1eqc at w=0 is 1.591 × rho, 1.06 × the 1.5 floor; priced (w=1) it is 2.73 × because the per-step canonicalisation/orientation work is charged like rho's. An earlier draft of the Iso Lab replay report (PR #3235) quoted 1.06 × for res1eq at w=1; that came from an approximation that left the IC arms' uncharged counters out and is wrong. The numbers above are the harness band.

Controls and caveats

- All 4608 records verified; `ecbench verify --replay 12`: audit OK, 12 replays reproduced, 0 problems.
- The A/A arm res1eqaa has the same method and seeds as res1eq, so it is bit-identical (ratio exactly 1.0000, degenerate interval); it checks determinism, not sampling noise. The sampling noise is the width of the intervals above.
- Local L0 run; counts are deterministic, no wall-clock claim. Class: engineering of a generic collision search.
- Remaining headroom to the floor is about 6% and below this design's resolution; this design cannot reach parity.

Evidence: `research/ic_res1eq_chain_20261101/` (PROTOCOL.md, specs-local, sessions/E-p36, analysis/).